Slurm plays a foundational role across supercomputing centres, AI training clusters, and research and enterprise-scale compute environments. Preserving its openness maintains its function as critical infrastructure glue for the global HPC and AI community, while reinforcing trust among developers, researchers, and operators.

The acquisition aligns with NVIDIA's broader expansion of open source and open AI initiatives. In early December, the company launched Alpamayo-R1, an open source reasoning vision-language model for autonomous driving research. NVIDIA also released new workflows and guidance for its Cosmos world models, which are licensed under a permissive open source framework to support the development of physical AI systems, including robotics and embodied intelligence.

Google and Meta bet on TorchTPU to break NVIDIA's hold over AI computing

Google is mounting an open source-led challenge to NVIDIA's dominance in AI computing by re-engineering its Tensor Processing Units (TPUs) to run PyTorch

efficiently, targeting the software layer that determines developer adoption at scale.

The effort, internally known as TorchTPU, aims to make Google's AI chips fully compatible with PyTorch, the world's most widely used open source framework for building and running AI models. By reducing friction for developers accustomed to NVIDIA GPUs, Google is seeking to lower the switching costs that have long locked the AI ecosystem into NVIDIA's CUDA software stack.

Google is also considering open sourcing parts of the TorchTPU software stack to accelerate adoption, addressing a key bottleneck that has limited TPU uptake despite competitive hardware performance. The move reflects a strategic shift away from relying on proprietary frameworks towards aligning with the tools most developers already use.

PyTorch, heavily supported by Meta Platforms, has become the default abstraction layer for AI development, with most engineers relying on its libraries rather than writing chip-specific code. Any AI accelerator that fails to run PyTorch efficiently faces structural resistance, regardless of raw compute capability.

NVIDIA's advantage stems not only from highly optimised GPUs but from CUDA's deep integration into PyTorch, built up over years of performance tuning. By contrast, Google's TPUs have historically been optimised for its internal Jax framework and XLA compiler, creating a mismatch with external developer workflows.

The initiative is commercially significant for Google Cloud, where TPUs are emerging as a key revenue driver. Google is working closely with Meta to improve PyTorch-on-TPU performance, aligning with Meta's goal of lowering inference costs and diversifying away from NVIDIA GPUs.

NVIDIA releases open source AI models for autonomous vehicles

NVIDIA has unveiled Alpamayo, a family of open source artificial intelligence models for autonomous vehicles, at CES 2026, positioning the platform as a direct challenge to proprietary systems such as Tesla's Full Self-Driving (FSD).

Designed to help vehicles "understand, reason and act in the real world," Alpamayo is a vision-language-action reasoning model built for Level 4 autonomy. Its core focus is addressing the industry's most difficult challenge: handling rare, unpredictable real-world scenarios while explaining driving decisions in a transparent and verifiable manner.

As a key differentiator, NVIDIA is open sourcing Alpamayo's core model, making it available on platforms such as Hugging Face. The company is also releasing more than 1,700 hours of real-world driving data and AlpaSim, an open simulation framework designed to stress-test autonomous driving stacks before deployment on public roads. NVIDIA says the open approach allows automakers and developers to customise, extend and validate autonomy systems using shared tools and datasets.

NVIDIA CEO Jensen Huang described Alpamayo as "the ChatGPT moment for physical AI," adding that robotaxis will be among the first to benefit from the platform as it moves into production use.

Tesla's Elon Musk acknowledged the significance of NVIDIA's move, stating on X that "it's easy to get to 99% and then super hard to solve the long tail of the distribution." Tesla's head of AI Ashok Elluswamy echoed the challenge, noting that "the long tail is sooo long, that most people can't grasp it."

Industry analysts argue that Alpamayo's open source design accelerates industry-wide innovation, while NVIDIA's shift towards a shared physical-AI ecosystem positions openness, explainability and simulation as critical advantages in gaining regulatory and public trust.

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